INTERNSHIP PROJECT

Telco Customer Churn Dataset Documentation

Introduction:

Customer churn refers to the situation where customers stop using a company's services. In the telecommunications industry, retaining customers is very important because acquiring new customers is more expensive than retaining existing ones.

This project focuses on analyzing customer behavior using the **Telco Customer Churn dataset** to predict whether a customer will leave the service or continue using it. Machine Learning techniques are applied to identify patterns and factors that influence customer churn.

The dataset contains information about customers such as demographics, services subscribed, billing information, and contract details. By applying data analysis and predictive modeling, the system helps telecom companies identify customers who are likely to churn and take preventive actions.

Data collection:

Source of data:

The dataset is publicly available and commonly used for machine learning projects related to customer retention.

Data description:

The dataset contains information about telecom customers including:

- Customer ID
- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Phone Service
- Internet Service
- Online Security
- Tech Support
- Streaming Services
- Contract Type
- Payment Method

- Monthly Charges
- Total Charges
- Churn Status (Target Variable)

Dataset size:

- Total Records: **7043 customers**
- Total Features: **21 attributes**
- Target Variable: **Churn (Yes/No)**

Data processing:

Data processing is an important step in preparing the dataset for machine learning models.

Data Cleaning

The following preprocessing steps were applied:

- Removed unnecessary columns such as **Customer ID**
- Handled missing values in **Total Charges**
- Converted data types where necessary
- Checked for duplicate records

Handling Categorical Variables

Most features in the dataset are categorical. These variables were converted into numerical form using:

- **Label Encoding**
- **One-Hot Encoding**

Examples:

- Gender → Male/Female
- Contract → Month-to-month / One year / Two year
- Payment Method categories

Feature Scaling

Numerical variables like:

- Tenure
- Monthly Charges
- Total Charges

were scaled using **Standard Scaler or Min Max Scaler** to improve model performance.

Data Splitting

The dataset was divided into:

- **Training Data (80%)**
- **Testing Data (20%)**

This helps evaluate the performance of machine learning models on unseen data.

Model building:

Several machine learning algorithms were used to build churn prediction models.

Models Implemented

1. **Logistic Regression**
    - Used for binary classification problems
    - Provides probability of churn
2. **Decision Tree Classifier**
    - Tree-based model
    - Helps understand important decision rules
3. **Random Forest Classifier**
    - Ensemble learning method
    - Combines multiple decision trees for better accuracy

Model Evaluation Metrics

Models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score

- Confusion Matrix

These metrics help determine how well the model predicts customer churn.

Hyperparameter Tuning:

Hyperparameter tuning improves the performance of machine learning models by selecting the best model parameters.

Techniques Used

Grid Search

Grid Search CV tests multiple combinations of parameters to find the best one.

Optuna (Advanced Optimization)

Optuna is used for efficient hyperparameter optimization by automatically selecting the best parameter combinations.

Example Parameters Tuned

For **Random Forest**

- Number of estimators
- Maximum depth
- Minimum samples split
- Minimum samples leaf

For **Decision Tree**

- Max depth
- Criterion
- Splitter

Hyperparameter tuning helps improve model accuracy and reduce overfitting.

Conclusion:

This project successfully analyzed the **Telco Customer Churn dataset** and built machine learning models to predict customer churn.

Key findings include:

- Customers with **month-to-month contracts** are more likely to churn.
- Higher **monthly charges** increase churn probability.
- Customers without **technical support or online security services** show higher churn rates.

Among the models tested, **Random Forest provided the best performance** due to its ensemble learning capability and robustness.

Machine learning models can help telecom companies identify high-risk customers and implement retention strategies.

Future Enhancements:

The project can be further improved with the following enhancements:

Deep Learning Models

Implement neural networks such as:

- Artificial Neural Networks (ANN)
- Deep Learning models using TensorFlow or Py Torch.

2. Real-Time Prediction System

Develop a **real-time churn prediction system** integrated with company databases.

3. Customer Segmentation

Apply **clustering techniques** to group customers based on behavior.

4. Advanced Feature Engineering

Create new features such as:

- Average spending
- Service usage score

- Customer engagement index

5. Deployment

Deploy the model using:

- **Flask / Fast API**
- **Stream lit Dashboard**
- **ML flow Model Tracking**

This will allow businesses to use the churn prediction system in real-time decision-maki